
Certified Biomedical Engineer (OIQ #5095040) with a concentration in Project Management from Polytechnique Montréal. I offer a diverse technical background spanning medical robotics, automated clinical systems, and FinTech applications. Passionate about software integrity and the IT industry, I specialize in developing rigorous quality specifications and automated testing frameworks. My professional philosophy is rooted in precision and efficiency: delivering innovative, safe, and high-quality technological solutions correctly the first time.

Professional work experience

Software Developer In Test (April 2021 > Today) | *National Bank of Canada*

- **Test Automation Engineering (Payment Hub & AEM):**
 - **Engineered** end-to-end automated functional and non-functional test suites in Java (Selenium and Cypress), significantly improving the reliability of the Payment Hub and the Bank's website redesign.
 - **Designed** Java-based automation for AEM components utilizing Gherkin/BDD methodologies to bridge the gap between technical requirements and business logic.
 - **Optimized** existing regression suites to reduce execution time, enhancing code maintainability and test suite scalability.
 - **AI-Enhanced Engineering:** Integrated GitHub Copilot into daily workflows to automate repetitive coding tasks, perform rapid prototyping of new test frameworks, and maintain high standards of code maintainability.
- **DevOps & CI/CD Integration:**
 - **Streamlined** software delivery by embedding automated tests into GitHub Actions pipelines, ensuring continuous quality checks.
 - **Orchestrated** CI/CD test cycles within Docker and OpenShift environments to simulate production-grade microservices deployment.
 - **Developed** robust retry and recovery testing logic for microservices using Cucumber to minimize false negatives in automated reports.
- **Leadership & Agile Collaboration:**
 - **Collaborated** with Product Owners and BI professionals to define technical acceptance criteria, ensuring 100% alignment with business objectives.
 - **Drove** code quality standards through rigorous GitHub code reviews, Pull Request (PR) approvals, and peer programming within the SDET team.
 - **Managed** the full defect lifecycle in Jira, from initial reporting through triage to final verification, ensuring zero-leakage of critical bugs.

Principal Software QA Analyst (June 2020 > April 2021) | *Logibec*

- **Test Strategy & Automation Engineering:**
 - **Architected** comprehensive test strategies and validation plans to achieve 100% functional coverage for mission-critical healthcare software.
 - **Engineered** multi-level automated suites (Unit, Integration, System) using Pytest and Katalon Studio, resulting in a significant reduction in manual regression overhead.
 - **Optimized** QA infrastructure by managing complex test environments, automated data preparation, and system-wide installations.
- **Cross-Functional Leadership & DevOps:**
 - **Directed** international knowledge transfer initiatives, aligning technical standards between local teams and offshore consultants in India to ensure global project consistency.
 - **Orchestrated** the full defect lifecycle, streamlining the feedback loop between developers and Scrum teams to eliminate bottlenecks and accelerate fix prioritization.
 - **Championed** CI/CD integration, facilitating the deployment of automated pipelines to accelerate release cycles and improve deployment frequency.
- **Governance, Standards & Risk Management**

- **Ensured** strict adherence to SDLC protocols, guaranteeing that product development met established regulatory and organizational benchmarks.
- **Spearheaded** risk management initiatives, leading meetings to identify, manage, and mitigate software risks throughout the project lifecycle.
- **Served** as a Subject Matter Expert (SME), delivering professional presentations on QA methodologies, defect prioritization, and advanced problem-solving techniques.

Quality Specialist (April 2019 > April 2020) | *Novo Électronique*

- **Regulatory Compliance & Certification:**
 - **Spearheaded** global product certifications (EMI/EMC, FCC, Wi-Fi, Bluetooth) and engineered compliance frameworks for ISO 13485 and ISO 14971 standards.
 - **Orchestrated** QMS implementation, authoring the Quality Manual, SOPs, and Work Instructions to ensure audit-readiness and organizational excellence.
 - **Managed** the full audit lifecycle, proactively resolving non-conformities through rapid containment and CAPA (Corrective and Preventive Action) management.
- **Systems Verification & Validation (V&V)**
 - **Led** cross-functional V&V teams (Software, Firmware, Hardware, Mechanical) to execute complex integration and regression testing for regulated medical devices.
 - **Safeguarded** patient safety by auditing the traceability of User Needs and Design Inputs against technical Risk Control Measures.
 - **Managed** Engineering Change Requests (ECR) for released products, conducting meticulous impact analyses to maintain system integrity.
 - **Authored** high-level Validation Reports and Test Plans, ensuring all clinical and technical requirements were validated prior to market release.

Quality Engineer (May 2018 > March 2019) | *Kinova Robotics*

- **Robotics Quality Engineering:**
 - **Orchestrated** the Quality Lifecycle for advanced robotic systems, ensuring strict alignment between engineering innovation and global regulatory mandates (CE, EMI/EMC, FCC).
 - **Spearheaded** Risk Management initiatives, utilizing Hazard Analysis, dFMEA, and pFMEA to mitigate critical failure modes during design and production.
 - **Served** as Subject Matter Expert (SME) for safety standards, including IEC 60601-1, ISO 13485, and ISO 14971.
- **Global Supply Chain & QMS Governance:**
 - **Managed** global supplier qualification and auditing, resolving complex quality bottlenecks through SCAR management and rigorous material inspections.
 - **Enforced** QMS integrity by managing CAPA, Non-Conformity reports, and Change Control procedures across multidisciplinary teams.
 - **Pioneered** synchronized quality protocols between Manufacturing Engineering and ERP systems to standardize production excellence.

Certifications

AWS (Amazon Web Services) | Certified Developer Associate (Valid February 25th 2023, expires on February 25th 2029)

AWS (Amazon Web Services) | Certified Cloud Practitioner (Valid August 15th 2023, expires on February 25th 2029)

ISTQB | Certified software tester (Obtained March 13th 2021)

Ordre des ingénieurs du Québec (OIQ) | Engineering permit – Membre#: 5095040

Internship Experience

System Design Intern in CAS R&D (January 2017 - May 2017) – Zimmer Biomet

- Regulatory Compliance: Authored and maintained critical quality documentation for Patient Specific Instrumentation (PSI), ensuring the integrity of Design History Files (DHF) and Device Master Records (DMR).
- Quality Engineering: Enforced rigorous safety and efficacy standards throughout the R&D lifecycle to ensure total product compliance.
- Software V&V: Developed comprehensive testing protocols and spearheaded the Verification and Validation (V&V) of emerging software through meticulous execution and reporting.

Client Technical Support intern (Jan 2016 – Sept 2016) – Dassault Systèmes

- Technical Problem Solving: Analyzed and resolved high-level technical issues for enterprise clients, maintaining end-to-end communication to ensure customer satisfaction.
- Software Specialization: Provided advanced technical support for the 3DEXPERIENCE platform, including CATIA, ENOVIA, and DELMIA.
- PLM Governance: Managed client workflows to ensure strict adherence to Product Lifecycle Management (PLM) protocols.

Production coordinator (Aug 2015 – Jan 2016) – V2R Biomedical

- Clinical Consultation: Collaborated with dental professionals to optimize patient treatment plans and prosthetic implant positioning.
- Process Automation: Engineered an automation system to accelerate 3D planning documentation and the generation of surgical jig models.
- CAD/CAM Modeling: Managed virtual treatment planning and high-precision CAD modeling for surgical applications.

Education: [Bachelor's degree in Biomedical Engineering \(Started Sept 2013 – Graduated May 2018\)](#)

Polytechnic Montreal - Concentration in project management

Relevant Skills: C/C++ SQL Python Java Cucumber HTML/CSS/JavaScript AGILE SDLC

Languages: [English:](#) Native proficiency [French:](#) Native proficiency

Interests and passions :

[Sports](#) – Ice Hockey, Tennis, Skiing, Gym, going on hikes

[Miscellaneous](#) – Travelling, playing the guitar, learning languages, dancing
